

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP40024
Subject Title	Microfabrication
Credit Value	3
Level	4
Pre-requisite / Co-requisite/ Exclusion	AP30020/ AP40006
Objectives	This course provides an introduction to the principles of microfabrication technologies. Key topics covered include photolithography, silicon etching, thin film deposition and etching, electroplating, micromachining, and bonding techniques.
Intended Learning Outcomes	Upon completion of this course, students will be able to: (a) classify various fabrication methods; (b) apply different fabrication steps based on the materials involved; (c) organize the process flow according to material compatibility and system design; and (d) develop and propose a fabrication sequence or method for a device/system.
Subject Synopsis/ Indicative Syllabus	Review of Basic Semiconductor Physics: principle of semiconductors materials and devices. Introduction to Microfabrication: material for fabrication, fabrication equipment, growth technology, and silicon-based process. Fabrication Technologies: pattern transfer technology, lithography – photolithography, resist technology, micromachining – wet etch and dry etch, materials processing technology, material deposition methods, basic ion implantation and diffusion doping process, low and high temperature process as well as device packaging methods Characterization technology for microfabrication processes: optical microscopy, scanning electron microscopy, transmission electron microscopy, atomic force microcopy, X-ray diffraction, energy dispersive X-ray spectroscopy, etc.
Teaching/Learning Methodology	Lecture: The design and fabrication of microelectronic devices and systems will be explained. Assignment sets will be given to assess the learning progress of students. Tutorial: Students will work on a set of problems in tutorials. Students are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance. These problem sets provide them opportunities to apply their knowledge gained from the lecture. They also help students to consolidate what they have learned. Furthermore, students can develop a deeper understanding of the subject in relation to daily life phenomena or experience.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting				
			a	b	c	d
	1. Continuous Assignments	40%	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓
	Total	100 %				
Continuous Assignments: Students will be given assignments and tests (quizzes) about the fundamental knowledge and key principles of various thin-film deposition techniques, materials characterization, microfabrication techniques, and device encapsulation. These will assess the knowledge of this subject achieved by the students.						
Examination: The end-of-semester final examination will cover all of the key issues related to the learning outcomes.						
Student Study Effort Expected	Class contact:					
	▪ Lecture & Tutorial				39 Hrs.	
	Other student study efforts:					
	▪ Reading and self-study				81 Hrs.	
	The total student study effort				120 Hrs.	
Reading List and References	Sami Franssila, Introduction to Microfabrication, 2 nd Edition, Wiley; 2010.					
	Marc J. Madou, Fundamentals of Microfabrication and Nanotechnology, 3 rd Edition, CRC Press, 2018.					
	Eun Sok Kim, Fundamentals of Microelectromechanical Systems (MEMS), 1st Edition, McGraw Hill, 2021.					
	Hwaiyu Geng, Semiconductor Manufacturing Handbook, 2 nd Edition, McGraw Hill 2018.					